Farjana Banna

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Education

University of Idaho *Moscow, Idaho, USA*
Ph.D. in Biological Sciences - Microbiology, Molecular Biology and Biochemistry (MMBB) *August 2026 - Present*
Advisor: Dr. Scott Grieshaber

Islamic University, Bangladesh *Kushtia, Bangladesh*
B.Sc in Biotechnology and Genetic Engineering *January 2019 - June 2024*
CGPA: 3.73/4.00 | Rank: 7th of 42 students

Research Experience

- **Graduate Researcher** *August 2026 - Present*
  Department of Biological Sciences, University of Idaho
  ◦ Investigating the molecular mechanisms that regulate the developmental cycle of Chlamydia trachomatis, with a focus on transitions between the infectious elementary body (EB) and replicative reticulate body (RB) forms.
  ◦ Examining heterogeneity in C. trachomatis developmental progression at the single-cell level to better understand how individual bacteria coordinate differentiation during intracellular infection.
  ◦ Developing experimental approaches using stage-specific markers and gene-expression analyses to characterize the timing and regulation of chlamydial developmental transitions.
  ◦ Investigating the roles of key developmental regulators, including Euo, HctA, and HctB, in controlling chlamydial differentiation and the production of infectious progeny.

- **Visiting Researcher** *January 2025 - January 2026*
  Aquatic Molecular Physiology Lab, Washington State University
  ◦ **Molecular Cloning:** Assisted in the development and preparation of **Tol2 plasmid construct** for stable transgenesis in rainbow trout, supporting efforts to accelerate muscle-related research.
  ◦ **Gene Validation:** Conducted routine **fluorescence screening** of transgenic fish embryos using high-resolution microscopy to validate successful integration and expression of the genetic construct.
  ◦ **Model Organism Handling:** Gained proficiency in handling and working with fish model organisms (trout) and experimental preparation.
  ◦ **Aseptic Lab Management:** Managed essential molecular biology procedures, including large-scale reagent preparation, sample archiving, and maintaining laboratory equipment.

- **Undergraduate Thesis Project** *July 2023 - May 2024*
  Islamic University, Bangladesh
  Biosynthesis and Characterization of Silver Nanoparticles (AgNPs) Using *Codiaeum variegatum* Leaf Extract and Evaluation of Antimicrobial Activity Against Antibiotic-Resistant Bacteria Isolated From Respiratory Diseases.
  ◦ **Novel Synthesis:** Synthesized and characterized **Silver Nanoparticles (AgNPs)** using *Codiaeum variegatum* leaf extract, confirmed by UV-vis spectrometric analysis.
  ◦ **Antimicrobial Efficacy:** Identified and isolated clinical **multidrug-resistant (MDR)** bacterial strains from human respiratory diseases and demonstrated the significant antibacterial activity of the biosynthesized AgNP against the isolated MDR strains, providing a potential green alternative therapeutic.

- **Research Group Member** *February 2022 - May 2024*
  Medical and Environmental Biotechnology Laboratory, Islamic University
  ◦ **Applied Microbiology:** Isolated and cultured salt-tolerant rhizosphere bacteria from the Sundarbans mangrove soil samples.
  ◦ **Bionanotechnology:** Synthesized AgNPs using four different plant extracts (lemongrass, mango leaf, ivy gourd, and noonflower) to compare synthesis yield and activity.

- **Computational Research Member** *April 2021 - March 2023*
  Hexagon Research Community, Bangladesh
  ◦ Performed **computational molecular docking studies** to predict potential inhibitors of Protein Arginine Methyltransferase-2 (PRMT2), a target protein in carcinoma.

Publications & Presentations

• **Publication:** MS Hossen, MN Islam, MEA Pramanik, **Farjana Banna** *et al.*, "Molecular Characterization and Potential Inhibitors Prediction of Protein Arginine Methyltransferase-2 in Carcinoma: An Insight from Molecular Docking, ADMET Profiling and Molecular Dynamics Simulation Studies." *Euroasian Journal of Hepato-Gastroenterology,* doi: 10.5005/jp-journals-10018-1443.

• **Poster Presentation:** Md Ashikuzzaman, Masuma Anzuman, **Farjana Banna** *et al.*, "Green Synthesis of Silver Nanoparticles Using *Coccinia grandis* Leaf Extract and Evaluation of its Antibacterial Activity against Multiple Drug-Resistant Fecal Pathogens." *5th International Conference on Biotechnology in Health and Agriculture (ICBHA),* September 2023.

Lab Expertise and Softwares

- **Experimental Expertise**:
  Tol2 Transgenesis System, Fluorescence Microscopy, PCR, Western Blotting, DNA/RNA Extraction (CTAB), Gel Electrophoresis, SDS-PAGE, Microbial Culture, MIC, Antibiotic Susceptibility Testing, Gram Staining, Disk Diffusion Method, UV Spectrophotometry, Silver Nanoparticle (AgNPs) Synthesis from leaf extract.
- **Computational Expertise**:
  BLAST, Multiple Sequence Alignment, Phylogeny, Protein-Protein Interaction (PPI), Molecular Docking (Pyrx, Pymol).
- **Tools and Software**:
  CodonCode Aligner, Swiss PDB Viewer, Mendeley, EndNote, Microsoft Office Suit, G Suit

Research-Oriented Industrial Visits

- **Carew & Co Bangladesh Ltd** *May 02, 2024*
  Darshana, Chuadanga, Bangladesh
  ◦ Visited their yeast culture laboratory.
  ◦ Witnessed the distillation process through fermentation, gaining insights into its role in production and product quality.

- **FnF Pharmaceuticals Ltd** *November 11, 2023*
  Jhenidah, Bangladesh
  ◦ Developed a comprehensive understanding of the production process for various animal vaccines, including manufacturing methods and quality control protocols.
  ◦ Visited the lyophilizer room, observing the freeze-drying process used to stabilize vaccines, with temperatures ranging from -40°C to 30°C.
  ◦ Explored the animal vaccine storage facility where manufactured animal vaccines are securely stored.

- **National Institute of Biotechnology (NIB)** *November 07, 2022*
  Dhaka, Bangladesh
  ◦ Explored the seven divisions of NIB: Molecular Biotechnology, Microbial Biotechnology, Plant Biotechnology, Environmental Biotechnology, Animal Biotechnology, Fisheries Biotechnology, and Bioinformatics.
  ◦ Gained insights into ongoing research projects across each division, enhancing understanding of cutting-edge biotechnological advancements.

Affiliation

- Member, Global Network for Bangladeshi Biotechnologist *November 2024 - Present*
- Member, American Society of Microbiology - Student Chapter, IU *May 2024 - Present*
- Volunteer, 37th Annual Conference of Bangladesh Society of Microbiologist (BSM) *January 26 - 27, 2024*
- Member, IAAS Bangladesh - IU Chapter *October 2020 - June 2024*
- Volunteer, Green Voice - IU Chapter *August 2020 - June 2024*
- Executive Member, Biotech Language Club - IU *August 2020 - June 2024*
- Volunteer, Cancer Awareness Program (CAP) *May 2019 - May 2024*
- Communication and Writing Secretary, Network of Young Biotechnologist of Bangladesh (NYBB) - IU Student Chapter *February 2021 - February 2024*

Honors and Awards

- National Science and Technology (NST) Fellowship *June 2024*
- Bangladesh Agricultural Olympiad 2020, Winner at 1st Round in Biotechnology and Genetics Category *February 2020*
- National Education Board Scholarship, Secondary School Certificate (SSC) Examination *2015*
- National Education Board Scholarship, Junior School Certificate (JSC) Examination *2012*

Selective Certifications

- **Impact Entrepreneureship Enabling Sustainable Future** *May 2024*
  ICT Division Bangladesh
- **Virtual Training on Computer Aided Drug Design for Beginners** *August 2021*
  Rajshahi University Science Club (RUSC)

Reference

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Dr. Michael Phelps
Assistant Professor
Department of Animal Sciences
Washington State University, Pullman, WA
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